

Bartik Instrument

Shift-share IV in trade economics

Imed Krisna Gupta

August 26, 2025

Gains exist, distribution is messy

- ▶ Aggregate gains from trade are robust across HO, Ricardian, and new-trade models.
- ▶ *Who* gains and loses depends on factor mobility.
- ▶ HO: gains accrue to the abundant factor (Stolper-Samuelson), *assuming factors move freely across industries*.
- ▶ SFM (Specific Factor Model): in the short run, some factors are tied to a sector → clear winners and losers.

The textbook distinction in one line

- ▶ HO: long run, perfect mobility, factor-price equalization.

The textbook distinction in one line

- ▶ HO: long run, perfect mobility, factor-price equalization.
- ▶ SFM: short run, sector-specific factors, sectoral wage gaps.

The textbook distinction in one line

- ▶ HO: long run, perfect mobility, factor-price equalization.
- ▶ SFM: short run, sector-specific factors, sectoral wage gaps.
- ▶ **Empirical reality:** the “short run” can last a decade or more, and the location of immobile factors is *regional*.

The textbook distinction in one line

- ▶ HO: long run, perfect mobility, factor-price equalization.
- ▶ SFM: short run, sector-specific factors, sectoral wage gaps.
- ▶ **Empirical reality:** the “short run” can last a decade or more, and the location of immobile factors is *regional*.
- ▶ This is what shift-share designs let us see in the data.

The within-country problem

- ▶ Cross-country evidence on gains from trade is settled.
- ▶ *Within*-country evidence is mixed and politically explosive.
- ▶ Why? Adjustment is **spatial**: districts, commuting zones, and provinces specialize in different sectors and absorb shocks differently.
- ▶ Reasons for imperfect mobility:
 - ▶ skill mismatch (sector-specific human capital)
 - ▶ geographic immobility (housing, family ties, language)
 - ▶ institutional frictions (labor laws, licensing)

What we want to estimate

We want a local causal effect of a trade shock:

$$Y_\ell = \alpha + \delta X_\ell + \varepsilon_\ell$$

- ▶ Y_ℓ — outcome in region ℓ (employment, wages, poverty)
- ▶ X_ℓ — local exposure to the trade shock (e.g., change in import penetration)
- ▶ δ — the parameter we care about

Problem. X_ℓ is endogenous: regions that import more are also regions where productivity, demand, or politics differ in unobserved ways.

→ We need an **instrument** for X_ℓ .

The Bartik (shift-share) idea

Two ingredients:

- ▶ **Shares** $s_{\ell k}$ — region ℓ 's pre-period exposure to sector k (employment, output, or expenditure share)
- ▶ **Shifts** g_k — a common (national or world) shock to sector k

Combine into a single regional exposure measure:

$$Z_{\ell} = \sum_k s_{\ell k} g_k$$

The instrument predicts *how hard* region ℓ should be hit by a common shock, given its industrial structure.

Why call it “Bartik”?

- ▶ Tim Bartik (1991) used national industry growth rates interacted with local industry mix to predict local labor demand growth.
- ▶ The construct predates him (Perloff 1957, Freeman 1980) but the name stuck after Blanchard & Katz (1992) popularized it.
- ▶ In trade: replace national growth with *tariff changes* (Topalova) or *import surges* (ADH).
- ▶ Goldsmith-Pinkham, Sorkin, and Swift (2020) formalize it as a GMM problem with the shares as instruments.

A first numerical look

3 regions $\times$ 2 sectors. Tariffs fall between 1987 and 1988.

region	L_{Agri}	L_{Manuf}	s_{Agri}	s_{Manuf}
A	100	9 000	0.011	0.989
B	1 000	400	0.714	0.286
C	700	3 000	0.189	0.811

National tariff change 1987 $\rightarrow$ 1988: Agri 30 $\rightarrow$ 10, Manuf 20 $\rightarrow$ 5.

Bartik exposure $Z_{\ell} = s_{\ell,\text{Agri}}\Delta \log t_{\text{Agri}} + s_{\ell,\text{Manuf}}\Delta \log t_{\text{Manuf}}$:

region	$Z_{\ell} (\Delta \log)$	$Z_{\ell} (\Delta \text{level})$
A	-1.38	-15.1
B	-1.18	-18.6
C	-1.33	-15.9

Note the **ranking flips**: log change makes A most exposed; level

What just happened?

- ▶ Manuf had the larger *proportional* cut: $\log(5/20) = -1.39$
vs. $\log(10/30) = -1.10$.

What just happened?

- ▶ Manuf had the larger *proportional* cut: $\log(5/20) = -1.39$ vs. $\log(10/30) = -1.10$.
- ▶ Agri had the larger *absolute* cut: -20 pp vs. -15 pp.

What just happened?

- ▶ Manuf had the larger *proportional* cut: $\log(5/20) = -1.39$ vs. $\log(10/30) = -1.10$.
- ▶ Agri had the larger *absolute* cut: -20 pp vs. -15 pp.
- ▶ Which one matters? Depends on the **outcome model**: log-log specifications need $\Delta \log t$, level specifications need Δt .

What just happened?

- ▶ Manuf had the larger *proportional* cut: $\log(5/20) = -1.39$ vs. $\log(10/30) = -1.10$.
- ▶ Agri had the larger *absolute* cut: -20 pp vs. -15 pp.
- ▶ Which one matters? Depends on the **outcome model**: log-log specifications need $\Delta \log t$, level specifications need Δt .
- ▶ Lesson: the Bartik is only as well-motivated as the production-side or trade-elasticity model behind it.

First stage, reduced form, IV

The Bartik is used as an instrument:

First stage $X_\ell = \pi_0 + \pi_1 Z_\ell + u_\ell$

Reduced form $Y_\ell = \rho_0 + \rho_1 Z_\ell + v_\ell$

2SLS $\hat{\delta}_{IV} = \rho_1 / \pi_1$

- ▶ Many papers (ADH 2013, Topalova) report the reduced-form coefficient ρ_1 directly, calling Z_ℓ “exposure.”
- ▶ The 2SLS scaling requires $\pi_1 \neq 0$ and the exclusion restriction $E[Z_\ell \varepsilon_\ell] = 0$.

Identification: two camps

Goldsmith-Pinkham, Sorkin, and Swift (2020) ask the central question: *which assumption makes the Bartik valid?*

Camp	Source of identification	Champion paper
Share-based	Shares $s_{\ell k}$ are exogenous to unobservables	Goldsmith-Pinkham, Sorkin & Swift (2020)
Shock-based	Shifts g_k are quasi-randomly assigned	Borusyak, Hull & Jaravel (2022)

Both deliver consistent IV under different conditions. The right diagnostic depends on which assumption you lean on.

Share exogeneity (GP framework)

Goldsmith-Pinkham, Sorkin, and Swift (2020) prove: the Bartik 2SLS estimator equals a GMM estimator where each **share** $s_{\ell k}$ acts as a separate instrument, weighted by **Rotemberg weights** α_k .

$$\hat{\delta}_{\text{Bartik}} = \sum_k \alpha_k \hat{\delta}_k$$

- ▶ $\hat{\delta}_k$ — just-identified IV using share $s_{\cdot k}$ alone
- ▶ α_k — Rotemberg weight, $\sum_k \alpha_k = 1$, can be negative
- ▶ A few industries usually carry most of the weight $\rightarrow$ the design hinges on the exogeneity of those few shares.

Rotemberg weights

For each industry k , the weight is

$$\alpha_k = \frac{g_k \cdot \sum_{\ell} s_{\ell k} X_{\ell}}{\sum_{k'} g_{k'} \cdot \sum_{\ell} s_{\ell k'} X_{\ell}}, \quad \sum_k \alpha_k = 1.$$

Two ingredients drive α_k :

- ▶ **size of the shift** g_k — industries with bigger national shocks matter more.
- ▶ **first-stage covariance** $\sum_{\ell} s_{\ell k} X_{\ell}$ — how strongly industry k 's shares correlate with the endogenous regressor.

Weights **can be negative**: if g_k and the first-stage covariance have opposite signs, that industry pulls the Bartik in the opposite direction. A handful of industries usually carry most of the weight.

Named for Julio Rotemberg (1983), who used the same decomposition logic in a different IV setting; Goldsmith-Pinkham, Sorkin, and Swift (2020) formalised it for the Bartik.

Rotemberg weights: 2x2

Use our toy data. Set $X_\ell = Z_\ell^{\text{lev}}$ so the mechanics are visible:

$X_A = -15.1$, $X_B = -18.6$, $X_C = -15.9$, with shifts

$g_{\text{Agri}} = -20$, $g_{\text{Manuf}} = -15$.

First-stage covariances:

$$\sum_\ell s_{\ell, \text{Agri}} X_\ell = 0.011(-15.1) + 0.714(-18.6) + 0.189(-15.9) = -16.45$$

$$\sum_\ell s_{\ell, \text{Manuf}} X_\ell = 0.989(-15.1) + 0.286(-18.6) + 0.811(-15.9) = -33.15$$

Numerators: $(-20)(-16.45) = 329.0$ and
 $(-15)(-33.15) = 497.2$. Denominator 826.2.

$\alpha_{\text{Agri}} = 0.40,$	$\alpha_{\text{Manuf}} = 0.60.$
--------------------------------	---------------------------------

Manufacturing carries 60% of the Bartik. With 100+ industries in Topalova or ADH, 5–10 industries typically carry 80% of the weight — those are the shares whose exogeneity you actually need to

Rotemberg weights in R

```
# shares: L × K matrix; g: length K; X: length L
fs_cov <- as.numeric(t(shares) %*% X)           # first-stage
alpha  <- g * fs_cov
alpha  <- alpha / sum(alpha)                   # Rotemberg

# per-industry just-identified IV
beta_k <- as.numeric(t(shares) %*% Y) / fs_cov
sum(alpha * beta_k)                           # = Bartik 2
```

The `bartik.weight` package (GitHub: paulgp/bartik-weight) does this and produces the Goldsmith-Pinkham, Sorkin, and Swift (2020) Table 5 diagnostic panel directly.

Diagnostics from Rotemberg weights

Goldsmith-Pinkham, Sorkin, and Swift (2020) recommend reporting:

1. **Concentration of weights** — Herfindahl of $|\alpha_k|$. If 2–3 industries dominate, your identification is really about those industries.
2. **Top-5 industries** — list them and ask: are their *shares* plausibly exogenous to local outcome trends?
3. **Pre-trend checks** on Z_ℓ — does pre-shock Z_ℓ predict pre-shock changes in Y_ℓ ?
4. **Just-identified IV by industry** — if $\hat{\delta}_k$ varies wildly across top industries, the pooled estimate is fragile.

Shock exogeneity (BHJ framework)

Borusyak, Hull, and Jaravel (2022) take a different route. Treat the **shifts** g_k as random and the shares as exposure weights.

- ▶ Identification: g_k is uncorrelated with sector-average residuals.
- ▶ Works well when shifts come from outside the system you study (e.g., supply-driven Chinese import surge from China's TFP and WTO accession).
- ▶ Diagnostic: tests for shock balance — does g_k correlate with pre-shock industry characteristics?
- ▶ The ADH “instrument with other rich countries' imports from China” exploits exactly this — it isolates the supply-driven component of the shock.

Inference: don't use OLS standard errors

Adão, Kolesár, and Morales (2019) show that conventional cluster-robust SEs **over-reject** with shift-share regressors:

- ▶ Residuals are correlated across regions that share industry mix.
- ▶ Their fix: an exposure-robust SE that treats *industries* as the clusters of randomness.
- ▶ In R, you can implement AKM SEs via the `ShiftShareSE` package or replicate by IV with industry-level data.
- ▶ Rule of thumb for class: if you see Bartik with white SEs only, be skeptical.

Limitations to keep in mind

- ▶ **Pre-period shares can be endogenous** (industries locate for a reason).

Limitations to keep in mind

- ▶ **Pre-period shares can be endogenous** (industries locate for a reason).
- ▶ **Common shocks may be heterogeneous** across sub-periods
→ time-varying g_k .

Limitations to keep in mind

- ▶ **Pre-period shares can be endogenous** (industries locate for a reason).
- ▶ **Common shocks may be heterogeneous** across sub-periods
→ time-varying g_k .
- ▶ **Placebo tests** on pre-period Y are essential.

Limitations to keep in mind

- ▶ **Pre-period shares can be endogenous** (industries locate for a reason).
- ▶ **Common shocks may be heterogeneous** across sub-periods → time-varying g_k .
- ▶ **Placebo tests** on pre-period Y are essential.
- ▶ **Weighting**: by region size, by inverse SE, by employment — affects which observations drive the result.

Limitations to keep in mind

- ▶ **Pre-period shares can be endogenous** (industries locate for a reason).
- ▶ **Common shocks may be heterogeneous** across sub-periods → time-varying g_k .
- ▶ **Placebo tests** on pre-period Y are essential.
- ▶ **Weighting**: by region size, by inverse SE, by employment — affects which observations drive the result.
- ▶ **Aggregation** matters: districts vs. provinces vs. commuting zones give different answers.

Topalova (2010): India 1991

- ▶ India's 1991 liberalization cut tariffs unevenly across industries.
- ▶ Cross-district variation in pre-reform industry mix → cross-district variation in exposure.
- ▶ Difference-in-differences with Bartik exposure as the treatment.

Identification: the *pace* of tariff cuts was dictated by external (IMF) pressure and was negotiated industry-by-industry without regard to district outcomes → shocks plausibly exogenous to district trends.

Topalova: instrument

$$Z_{\ell}^{\text{Top}} = \sum_k s_{\ell k}^{1987} \Delta \log(\text{Tariff}_k)$$

- ▶ $s_{\ell k}^{1987}$ — district ℓ 's pre-reform employment share in industry k
- ▶ $\Delta \log(\text{Tariff}_k)$ — national log-change in industry k 's tariff between 1987 and 1997

Outcomes: district-level poverty headcount, poverty gap, consumption growth.

Topalova: specification

Topalova runs a **panel** across four NSS rounds (1983, 1987–88, 1993–94, 1999–2000):

$$Y_{dt} = \alpha_d + \gamma_{st} + \delta \text{Tariff}_{dt} + \mathbf{W}'_{dt} \lambda + \varepsilon_{dt}$$

with district tariff constructed Bartik-style,

$$\text{Tariff}_{dt} = \sum_k s_{dk,1987} \cdot \text{Tariff}_{kt}$$

- ▶ α_d — district FE (kill all time-invariant district traits)
- ▶ γ_{st} — state $\times$ year FE (absorb state-wide trends)
- ▶ shares **frozen at 1987**; national tariffs Tariff_{kt} vary year-by-year
- ▶ IV: instrument *realised* tariffs with the *scheduled* tariff reduction path

Under two-way FE this is numerically equivalent to a long-difference regression on ΔY vs. ΔTariff — which is why ADH style papers

Topalova: findings

- ▶ Rural districts more exposed to tariff cuts → **slower decline in poverty**, lower consumption growth.
- ▶ Effect concentrated among the geographically immobile and least skilled.
- ▶ **Mechanism:** rigid labor laws → costly to fire and to expand → factor reallocation stalls.
- ▶ In states with flexible labor regulation, the poverty effect of exposure is statistically zero.

Takeaway: Bartik captures *exposure*; institutions decide *adjustment*.

Autor, Dorn & Hanson (2013): the China shock

- ▶ Autor, Dorn, and Hanson (2013) examine US commuting zones (CZs) 1990–2007.
- ▶ China's WTO accession (2001) and supply-side reforms drove a sustained import surge.
- ▶ Variation across CZs comes from pre-shock industry mix.

ADH: the instrument

$$\Delta IPW_{c\tau} = \sum_k \frac{L_{ck,t_0}}{L_{k,t_0}} \cdot \frac{\Delta M_{k\tau}^{US}}{L_{c,t_0}}$$

- ▶ L_{ck,t_0} — start-of-period employment in CZ c , industry k
- ▶ $\Delta M_{k\tau}^{US}$ — change in US imports from China in industry k

To address shock endogeneity (US demand could pull in imports too), ADH instrument ΔM^{US} with ΔM^{OTH} — imports from China to *eight other rich countries*. This isolates the China-side supply shock.

ADH: main results

- ▶ High-exposure CZs lose more manufacturing jobs.

ADH: main results

- ▶ High-exposure CZs lose more manufacturing jobs.
- ▶ Wages fall, esp. among non-college workers.

ADH: main results

- ▶ High-exposure CZs lose more manufacturing jobs.
- ▶ Wages fall, esp. among non-college workers.
- ▶ **No offsetting employment gains** in other sectors at the local level.

ADH: main results

- ▶ High-exposure CZs lose more manufacturing jobs.
- ▶ Wages fall, esp. among non-college workers.
- ▶ **No offsetting employment gains** in other sectors at the local level.
- ▶ Effects persist for at least a decade.

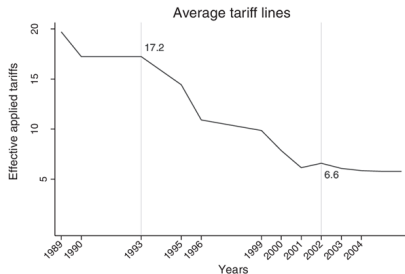
ADH: main results

- ▶ High-exposure CZs lose more manufacturing jobs.
- ▶ Wages fall, esp. among non-college workers.
- ▶ **No offsetting employment gains** in other sectors at the local level.
- ▶ Effects persist for at least a decade.
- ▶ Autor, Dorn, and Hanson (2016) (2016 review) extends to job churning, lifetime earnings, social spending, marriage rates, mortality.

Why ADH matters methodologically

- ▶ Cleanest example of the **shock-based** identification logic in BHJ.
- ▶ The “other rich countries” instrument is the canonical move for purging demand-side endogeneity from a Bartik.
- ▶ Sparked the AKM (2019) literature on inference because their headline SEs were too tight.
- ▶ Their data and Stata replication files are widely used in teaching.

Kis-Katos & Sparrow (2015): Indonesia 1993–2002



- ▶ 259 districts; outcome is district-level poverty.
- ▶ Two waves of tariff cuts: WTO 1995 + IMF program 1999.
- ▶ Innovation: separate **output** and **input** tariff exposure using a 1990 input-output table.

Output vs. input tariffs

$$Ot_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,t=0}}{Q_{k,t=0}} \times t_{st} \right)$$

$$It_{kt} = \sum_{s=1}^S \left(\frac{Q_{sk,t=0}}{Q_{k,t=0}} \times \sum_{j=1}^J \frac{M_{js,1990}}{M_{s,1990}} t_{jt} \right)$$

sector s , district k , time t , input sector j , labor Q , tariff t

The first share weights output tariffs by district industry mix; the second additionally weights through the I-O matrix to capture *imported intermediate inputs* embodied in each sector.

Output tariff exposure

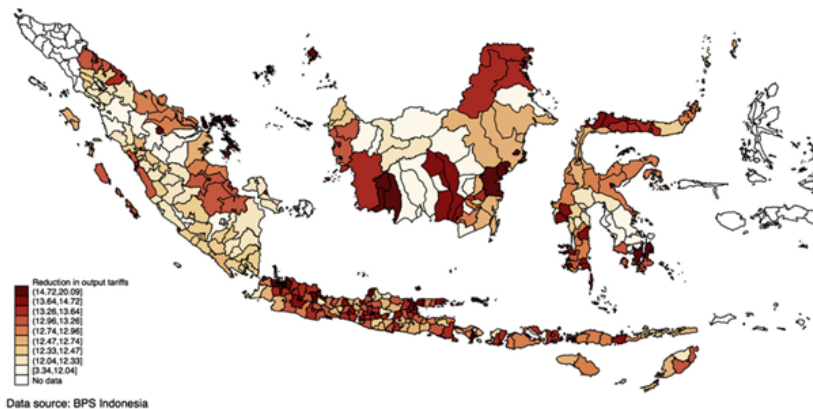
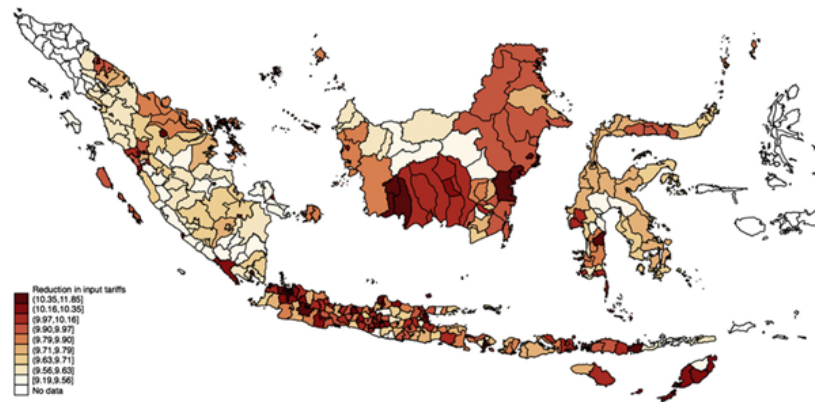


Fig. 3. Reductions in labor weighted district output tariff measures (1993-2002).

Input tariff exposure



Data source: BPS Indonesia

Fig. 4. Reductions in labor weighted district input tariff measures (1993-2002).

Kis-Katos & Sparrow: findings

$$\Delta y_{kt} = \alpha + \beta_1 Ot_{kt} + \beta_2 It_{kt} + \gamma \Delta X'_{kt} + I'_k \theta + \lambda_{rt} + \Delta \epsilon_{kt}$$

- ▶ $\beta_1 > 0$ — output tariff cuts **raise** poverty (import competition story).
- ▶ $\beta_2 < 0$ and **larger in magnitude** — input tariff cuts **reduce** poverty (cheaper intermediates $\rightarrow$ firm competitiveness $\rightarrow$ low-skill work participation and middle-skill wages rise).

Net effect on Indonesian poverty is negative — but only because input tariff liberalization dominates. The distributional story matters.

Three applications side by side

	Topalova (2010)	ADH (2013)	KK & Sparrow (2015)
Country	India	US	Indonesia
Episode	1991 reform	1990–2007	1993–2002
Unit	districts	commuting zones	districts (259)
Shock	tariff cuts	China imports	tariff cuts (2 waves)
Share	1987 employment	start-period emp.	I-O × employment
Outcome	poverty, consumption	jobs, wages	poverty
Direction	exposure ↑ → poverty ↑	exposure ↑ → jobs ↓	output↑ poverty↑, input↓ poverty↓

Hands-on with R

The dataset

dat.xlsx — 3 regions, 2 sectors, 2 years. Toy data so you can verify by hand.

```
library(readxl)
library(dplyr)
library(tidyr)

dat <- read_excel("dat.xlsx")
dat
```

A tibble: 12 × 5

	year	region	sector	tariff	labor
	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	1987	A	Agri	30	100
2	1987	B	Agri	30	1000
3	1987	C	Agri	30	700
4	1987	A	Manuf	20	9000
5	1987	B	Manuf	20	400
6	1987	C	Manuf	20	3000

Step 1 — compute 1987 employment shares

```
shares <- dat |>
  filter(year == 1987) |>
  group_by(region) |>
  mutate(share = labor / sum(labor)) |>
  select(region, sector, share)
```

shares

A tibble: 6 × 3

	region	sector	share
	<chr>	<chr>	<dbl>
1	A	Agri	0.0110
2	A	Manuf	0.989
3	B	Agri	0.714
4	B	Manuf	0.286
5	C	Agri	0.189
6	C	Manuf	0.811

Step 2 — compute national tariff shocks

```
shocks <- dat |>
  distinct(year, sector, tariff) |>
  arrange(sector, year) |>
  group_by(sector) |>
  summarise(
    dlog_t = log(last(tariff)) - log(first(tariff)),
    dlev_t = last(tariff) - first(tariff)
  )
shocks
```

```
# A tibble: 2 × 3
  sector dlog_t dlev_t
  <chr>   <dbl>  <dbl>
1 Agri    -1.10    -20
2 Manuf   -1.39    -15
```

Step 3 — assemble the Bartik

```
bartik <- shares |>
  left_join(shocks, by = "sector") |>
  group_by(region) |>
  summarise(
    Z_log = sum(share * dlog_t),
    Z_lev = sum(share * dlev_t)
  )
bartik
```

```
# A tibble: 3 × 3
  region  Z_log  Z_lev
<chr>   <dbl> <dbl>
1 A      -1.38 -15.1
2 B      -1.18 -18.6
3 C      -1.33 -15.9
```

Region A is most exposed in log, region B most exposed in levels — same point we made earlier, now in code.

Step 4 — the (would-be) 2SLS

With only 3 districts there's nothing to estimate. In a real Topalova-style exercise:

```
library(fixest)

# pretend `panel` has many districts × years with outcome `y`
# realised exposure `Xkt`, instrument `Zkt`, controls W, state fixed effects
m <- feols(
  dpov ~ W1 + W2 | state^year | Xkt ~ Zkt,
  data = panel,
  cluster = ~ industry_topshare
)
summary(m)
```

For AKM-corrected SEs, use the ShiftShareSE package and pass the share matrix.

Exercise (10 min, in pairs)

Using only `dat.xlsx`:

1. Recompute Z_ℓ assuming the 1988 shares (not 1987). What's different and why?
2. Now compute it using **employment growth** as the shift instead of tariffs. Does the ranking of regions change?
3. Region A is essentially a “manufacturing town.” If we drop A, what happens to the cross-sectional variation in Z ? What does that tell you about the Rotemberg-weight diagnostic in GP?
4. Suppose we observed poverty rates falling by 5pp in A, 2pp in B, 4pp in C. Compute the reduced-form slope of $\Delta\text{poverty}$ on Z_ℓ . Interpret with care — what assumption are you making?

What to read next

- ▶ Goldsmith-Pinkham, Sorkin, and Swift (2020) — the canonical methods paper. Tables 5–6 are the diagnostic template you should copy in your own work.
- ▶ Borusyak, Hull, and Jaravel (2022) — when shocks (not shares) carry identification.
- ▶ Adão, Kolesár, and Morales (2019) — the inference paper. The R package ShiftShareSE implements it.
- ▶ Autor, Dorn, and Hanson (2013) + Autor, Dorn, and Hanson (2016) (review) — the China shock canon.
- ▶ Topalova (2010) — the trade-and-poverty workhorse for developing countries.
- ▶ Kis-Katos and Sparrow (2015) — the closest Indonesian benchmark for your own work.

Frontier and Indonesian context

- ▶ I-O Bartik: extend KK & Sparrow's input/output split to services value-added (GVC trade).
- ▶ **Granular instruments:** large firms as shock units instead of sectors (Gabaix-Koijen 2024).
- ▶ **Uncertainty as a shift:** ongoing work by **Pane, Massie & Gupta (2025)** uses a geopolitical uncertainty index in place of tariff changes — the share captures exposure to *uncertain* trading partners rather than to specific industries (Pane, Massie, and Gupta 2025).
- ▶ Whatever the shift, the identification logic is the same: argue for either share exogeneity or shock exogeneity, then *show diagnostics consistent with that argument*.

Wrap-up

- ▶ Bartik = shares $\times$ shifts $\rightarrow$ predicted regional exposure.

Wrap-up

- ▶ $\text{Bartik} = \text{shares} \times \text{shifts} \rightarrow \text{predicted regional exposure}.$
- ▶ Two identification stories; pick one and live by it.

Wrap-up

- ▶ $\text{Bartik} = \text{shares} \times \text{shifts} \rightarrow \text{predicted regional exposure}.$
- ▶ Two identification stories; pick one and live by it.
- ▶ Always report Rotemberg weights or shock-balance tests.

Wrap-up

- ▶ $\text{Bartik} = \text{shares} \times \text{shifts} \rightarrow \text{predicted regional exposure}.$
- ▶ Two identification stories; pick one and live by it.
- ▶ Always report Rotemberg weights or shock-balance tests.
- ▶ Use AKM SEs, not white SEs.

Wrap-up

- ▶ $\text{Bartik} = \text{shares} \times \text{shifts} \rightarrow \text{predicted regional exposure}.$
- ▶ Two identification stories; pick one and live by it.
- ▶ Always report Rotemberg weights or shock-balance tests.
- ▶ Use AKM SEs, not white SEs.
- ▶ The trade-policy applications (Topalova, ADH, KK & Sparrow) give you a recipe you can transplant onto any Indonesian liberalization episode.

References

- Adão, Rodrigo, Michal Kolesár, and Eduardo Morales. 2019. "Shift-Share Designs: Theory and Inference." *Quarterly Journal of Economics* 134 (4): 1949–2010.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review* 103 (6): 2121–68.
- . 2016. "The China Shock: Learning from Labor-Market Adjustment to Large Changes in Trade." *Annual Review of Economics* 8: 205–40. <https://www.jstor.org/stable/26774368>.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2022. "Quasi-Experimental Shift-Share Research Designs." *Review of Economic Studies* 89 (1): 181–213.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *The American Economic Review* 110 (8): pp. 2586–2624.